EXP-002 Model Evaluation Report

Evaluation · MealSenseAI · Joselito Dizon · 2026-08-18

Developer-reported accuracy of the EXP-002 food classifier and the ONNX runtime parity check.

Model

EXP-002 is the food classifier used in the MealSenseAI Solution Architecture, trained on the merged Food-101 and FoodSG-233 taxonomy.

Results

Developer-reported results are 70.63% top-1 accuracy, 85.66% top-3 accuracy and 66.68% macro-F1.

Roughly three in ten photos get a wrong top-1 class, which is why the app always shows alternatives and asks the user to confirm.

Runtime parity

The ONNX export was checked against PyTorch on 601 images: top-1 and top-3 agreement was 100%.

Parity shows the runtime reproduces the checkpoint; it is not an accuracy measurement.